AOCSEQ AN R PACKAGE FOR MIXED CELL TYPE ANNOTATION

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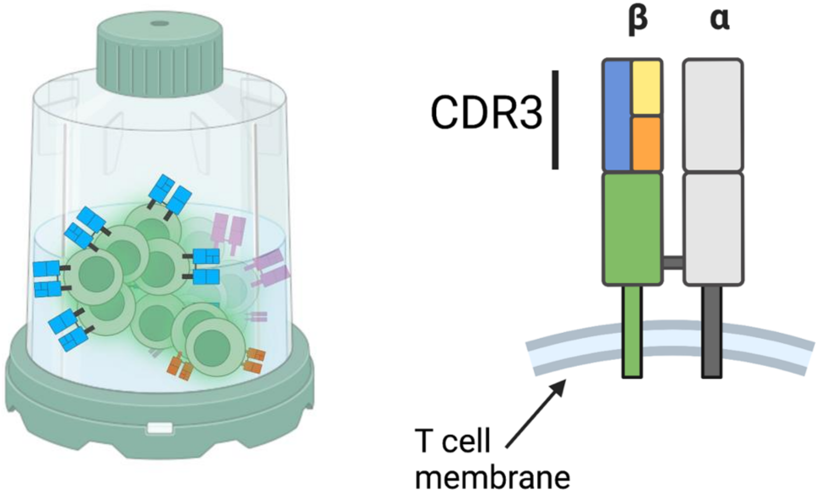
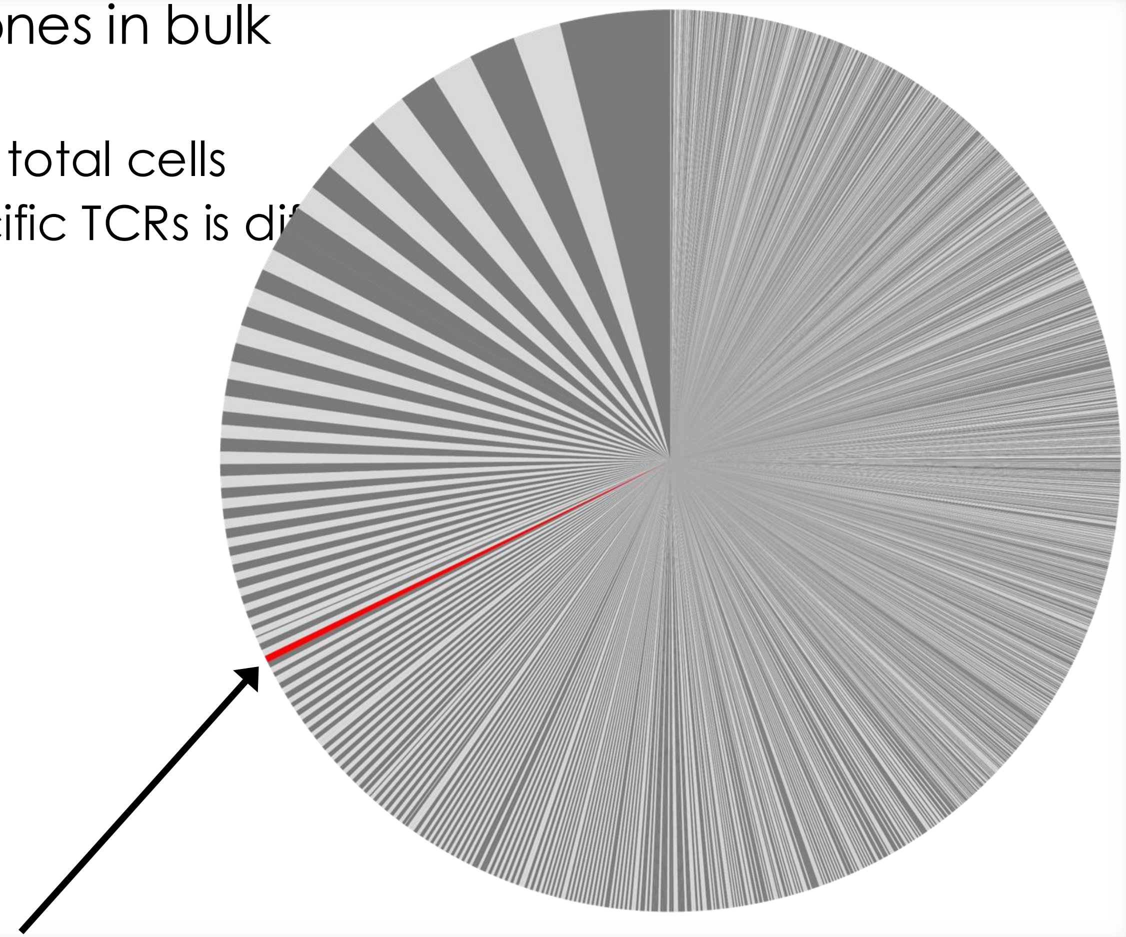
ISOLATING TCRS REMAINS PROBLEMATIC

TA specific TCR[illegible] KRAS/MAGEA4

- Low frequency of antigen-specific clones in bulk
  - Outgrowth of bystander T-cells
  - High TCR diversity ~10^8 unique in ~10^{11} total cells
  - Identification of antigen-specific specific TCRs is di

- Single cell RNA sequencing (scRNAseq)
  - Sequence the TCR and transcriptome for the mRNA of marker genes





Functional & high avidity TCR – G12C (0.23%, 9 cells. 2,300 in 10^6)

DIMENSION REDUCTION IS SUBJECTIVE

- Number of PCs included in information used for KNN graph is arbitrary
    - Based on industry standards – variable
    - No method to rank cells – problematic if enriching cell subsets

- Different parameter values produce different clusters

- Cell annotation and functional experiments
    - Software freely available for cell type annotation is lacking

PDB DOI: https://doi.org/10.2210/pdb7PB2/pdb

PURPOSE & AIMS

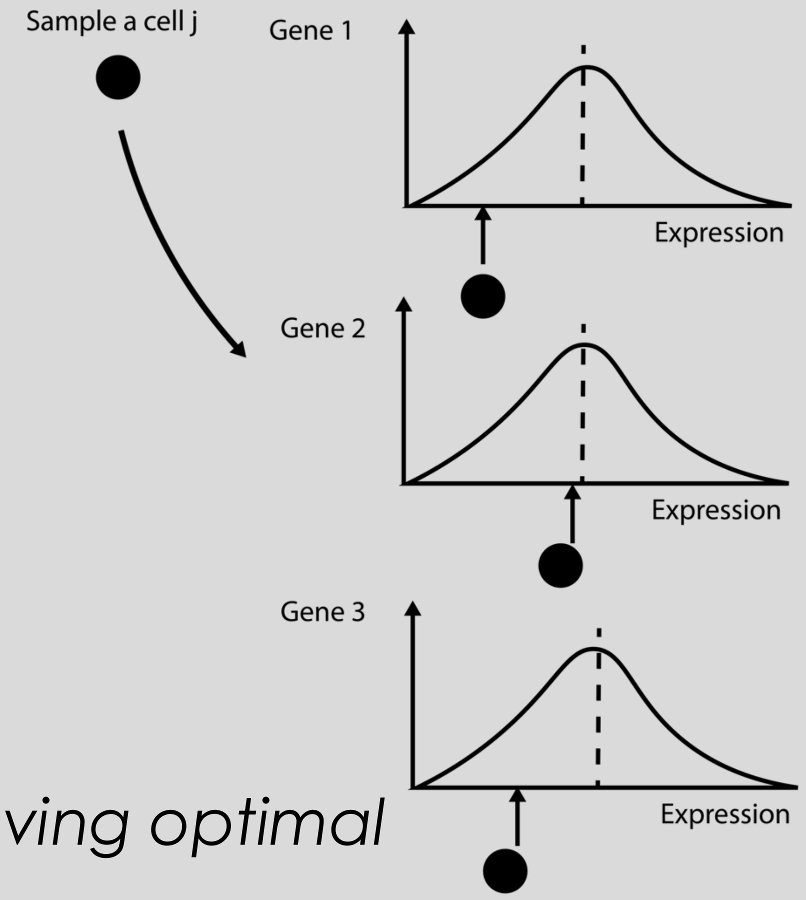
- Lack of mixed annotation software
  - Theoretical framework to rank cells.
  - Compare with granular approaches.
    - KNN graphs or community networks.

Example: TCR activation

- optimal characteristics.

- *The likelihood is the probability of a single cell having optimal characteristics.*
- *Using this approach, we can rank TCRs*[1]

Mahalanobis distance



$$d_M(\bar{x}, Q) = \sqrt{(\bar{x} - \bar{\mu})S^{-1}(\bar{x} - \bar{\mu})}$$

1 Woods ML, et al., ACS Biol. 2016 Jun 17;5(6):459-70.

SOFTWARE DESIGN

- Currently supports
  - 10X genomics cell ranger feature barcode matrices.
  - Adaptive immunosequencing data.

- Multimodal data
  - VDJ enrichment and immunoprofiling.
  - Multiplex data.
    - Hash tagged data.
    - Cell surface labelling.

- Extensions
  - Spatial RNA sequencing.
    - 10X genomics with integration of voyager for Geo spatial analysis

AOCSEQ – BETA RELEASE

- **1.0 Import:** "EM algorithm","GMM demux",**"CombineData"**
- **1.1 Annotation: "AnnotateClonotypes"**
- **2.0 Classification:** "DEsingle", **"ClonotypeClassification"**
- **3.0 Expression:** "GetSpecificCells", **"GetGeneSignature"**
- **3.1 References: "GetPerturbationReference"**
- **4.0 Distances: "ClassifyCells"**, "AddDistances"
- **4.1 Ranking: "PlotPosterior"**
- **5 Plotting: "segmentPlot"**, "AnnotateUMAP"

In development

- Spatial RNA sequencing.
  - Voyager integration for Geo spatial analysis.
- Other cell type annotation – NK and B cells.

1.0 DATA IMPORT & ANNOTATION.

- **CombineData:** - Demultiplexing and Quality control (QC).
  - Imports data
    - GMM demux[1] – EM algorithm & Gaussian mixture model.
  - Imports Seurat[2] QC for removal of outliers and filtering.
  - Annotates cells with metadata.
    - TCR sequences.
    - Marker gene of interest (goi) - High or unassigned expression.
    - Classifies T cells as CD4 or CD8 in phenotype.

1.1 ANNOTATION & CONSOLIDATION

- **Annotate Clonotypes:** – Summarization and consolidation.
  - Statistical ranking.
    - Rank cell types by the mean cell number in a group.
      - across multiple or replicate samples from a cell product.
  - Output: A summary data sheet.
    - Lists the percentage of cells in each cell type with high expression of each marker gene.
    - Lists associated metadata.

BAYESIAN INFERENCE & FITTING

- **ClonotypeClassification:** - Bayesian model fitting of zero-inflated count data (Noah Adds information on classification tree)

  - Imports a summary data sheet.
    - implements DEsingle on each ranked cell type listing p-values for differential significance tests.

  - Can be used with a reference and test or can compare each ranked cell type to pseudo-bulk sequencing.

  - Output: A gene classification data sheet.
    - Lists the p-value of cell types with significant differential status & abundance.

DE SINGLE

- **ClonotypeClassification:** - Bayesian model fitting of zero-inflated count data (Noah Adds information on classification tree)

  - Imports a summary data sheet.
    - implements DEsingle on each ranked cell type listing p-values for differential significance tests.

  - Can be used with a reference and test or can compare each ranked cell type to pseudo-bulk sequencing.

  - Output: A gene classification data sheet.
    - Lists the p-value of cell types with significant differential status & abundance.

OUTLIER CLASSIFICATION

- **ClonotypeClassification:** - Bayesian model fitting of zero-inflated count data (Noah Adds information on classification tree)

  - Imports a summary data sheet.
    - implements DEsingle on each ranked cell type listing p-values for differential significance tests.

  - Can be used with a reference and test or can compare each ranked cell type to pseudo-bulk sequencing.

  - Output: A gene classification data sheet.
    - Lists the p-value of cell types with significant differential status & abundance.

3.0 GENE EXPRESSION ANALYSIS

- GetGeneSignature: Differential gene expression analysis of cells grouped by differential abundance of marker genes.
  - Imports a gene classification data sheet.
    - Imports Seurat FindMarkers to identify differentially expressed genes.
  - Splits differential expression by-
    - Cell metadata.
  - Output: Differential expression data sheet.
    - Lists the results by cell meta data

3.1 NORMALIZING REFERENCE DATA

- GetPerturbationReference: Generates a normalized count matrix from cells with optimal gene expression.

  - Imports data that will be used as a reference.
    - Selects cells based on the expression of marker genes of interest.

  - Combines the normalized values of desirable cells to be included in the reference.

  - Output: A gene classification data sheet.
    - Lists the p-value of cell types with significant differential status & abundance.

ANALYSIS – 4.0 DISTANCES

- **ClassifyCells:** Computes the single cell distance of a test data set to a reference
- Imports a summary data sheet, a clonal object and reference
    - Implementation
    - Taxicab, Euclidean, z-score and Mahalanobis distance bubblesort algorithm for the matrix inversion.

- After cells have been classified, distances from reference can be added as metadata.

- Output: A distance data sheet.
    - Lists the mean distance from the reference for each cell type.

4.1 SINGLE CELL RANKING

- EstimatePosterior: Implements approximate Bayesian computation sequential Monte Carlo on the data.
  - Imports a summary data sheet, clonal object and reference dataset.
    - implements ABCSMC for each cell type, computing a distance from the prior multi-dimensional normal distribution.
  - Can be used to compute the robustness of a clonal repertoire to activation response.
  - Output: A marginal likelihood data sheet.

5 DATA VIZUALIZATION

- **Plotting: "segmentPlot", "AnnotateUMAP" , "Heatmap", "VolcanoPlot"**
  - Imports a summary data sheet or clonal object.
    - implements dimension reduction by importing Seurat functionality.
  - SegmentPlot imports Circos.R and can be used to compare the frequency of clonotypes between samples.
  - Output: A set of images generated by the plotting tools
    - Functionality for plotting includes a circos plot, Venn diagram,

VIZUALIZATION